Fast Pixel-Superpixel Region Merging for SAR Image Segmentation

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***Abstract*— In this article, we propose a fast superpixel region merging algorithm for synthetic aperture radar (SAR) image segmentation. With our previously proposed adaptive superpixel generation approach (ALFCE), an initial over-segmentation superpixel map for SAR imagery can be obtained. A sketch edge map is used here to eliminate the mixed superpixels to refine the over-segmentation. Then, we focus on rapid superpixel merging for efficient and accurate SAR image segmentation by using the statistical region merging (SRM) framework. This article proposes a new merging order with the consideration of statistical dissimilarity measure and common boundary length penalty, as well as the homogeneity constraint for each superpixel pair. For the merging predicate, we define an adaptive merging threshold according to the image complexity, making the proposed superpixel merging no need to set any merging parameters in advance. Disjoint set is utilized in this article to map the superpixel pairs to pixel pairs for the sake of fast region merging, which has a low computation cost even with the increasing of superpixels. Experimental results on synthetic and real SAR images demonstrate that the segmentation precision of our proposed method can reach more than 85% and also superior to other state-of-the-art methods in terms of computational efficiency.**

***Index Terms*— Disjoint set, dissimilarity measure, edge penalty, superpixel merging, synthetic aperture radar (SAR) image segmentation.**

I. Introduction

AS AN active imaging system operating in all weather and day–night conditions, synthetic aperture radar (SAR) has been widely used in many fields such as agriculture monitoring, land resources management, urban development, and military [1]. SAR image segmentation is a key step for remote sensing information extraction and interpretation, which partitions an SAR image into several nonoverlapping homogeneous segments, thus providing a region-based representation. However, the inherent speckle noise, serious geometric distortion, and complex backscattering within SAR images make the segmentation still very challenging [2]–[4].

Until now there have been proposed various SAR image segmentation approaches such as spectral iterative clustering [5], [6], active contour model [7]–[9], level set [10]–[12], graph cut [13], [14], edge-based scheme [3], [15]–[17], region-based scheme [18]–[22], and hybrid scheme [23]–[27]. Since edge- and region-based segmentation methods make use of not only the geometry structure information but also the spatial relationship among neighboring pixels within the SAR images, these methods can reduce the negative effect of speckle noise on image segmentation. More importantly, image details such as isolated targets and lines can be preserved to a large extent. Therefore, these methods and the hybrid scheme are popular in SAR image segmentation field. Edge-based segmentation scheme aims at finding edges in SAR images and thus region can be recovered from closed contours. Therefore, the segmentation performance highly depends upon edge detection stage, which requires a suitable edge detector for SAR images. To date, several edge detectors with constant false alarm rate (CFAR) [28]–[31] for SAR images based on the ratio of averages (ROAs) or the likelihood ratio are proposed. Zhang *et al.* [3] proposed to use the Bhattacharyya coefficient with rectangle-window configuration instead of the original ratio-based edge detector for SAR image edge detection. Then, the edge penalty is incorporated into merging cost for SAR image segmentation. Shui and Cheng [15], [30] proposed a new ratio-based edge detector with Gaussian–Gamma-shaped (GGS) windows configuration and then introduced it to watershed transform for the initial over-segmentation of SAR images. Yu and Clausi [16] proposed to use the graduated increased edge penalty within the Markov random field (MRF) context model for region growing to achieve SAR image segmentation. It is worth pointing out that due to the speckle noise and low gray level contrast in SAR images, the final edges always present many false edges or missing points, which may easily lead to incorrect segmentation.

Region-based segmentation scheme generally starts from initial over-segmentation and then achieves final image

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segmentation by region merging. Initial over-segmentation is usually generated by watershed transform [32] or superpixel segmentation methods [33]–[36]. Region merging is iteratively implemented in terms of merging costs, which usually can be defined on the basis of dissimilarity of two regions. Carvalho *et al.* [2] presented SAR image segmentation by using statistical region growing and hierarchical merging strategy. The algorithm first combines the coefficient of variation analysis with the region growing method to generate initial over-segmentation. Then, two most statistically similar regions are iteratively merged until a proper stop criterion is satisfied. Fan *et al.* [18] proposed to use multiscale Bhattacharyya distance to generate initial over-segmentation and then represent it using undirected region adjacent graph (RAG). The region-merging process is performed with a merging cost in consideration of statistical similarity measure and the relative common boundary length penalty (RCBLP). Yu *et al.* [20] proposed a region-based SAR image segmentation method where TurboPixels algorithm is utilized to generate the over-segmentation. Then, different features such as brightness, texture, edges, and spatial information are used to determine the merging cost, which is used to iteratively merge segments to complete the final segmentation. Xia *et al.* [37] proposed to use the MRF model on region adjacency graph to achieve SAR image segmentation. The initial over-segmentation map is obtained by the watershed transform algorithm. It should be noted that most of these mentioned methods simply use existing over-segmentation algorithms (e.g., watershed, TurboPixels algorithms) that are originally designed for natural images to generate initial partitions. Due to speckle noise in SAR images, these initial partitions may contain edges between different regions, which will lead to incorrect segmentation. Moreover, the computational costs of these methods are relatively high due to the iterative merging, especially when the number of initial partitions is large. Although Shui and Zhang [15] utilize the nearest neighbor graph (NNG) for fast search minimal edge on a region adjacency graph to accelerate the iterative region merging, the segmentation results can be easily affected by the termination iteration parameter, e.g., number of final regions.

In this article, we try to develop an efficient image segmentation algorithm for SAR images. Our proposed method belongs to the region-based segmentation scheme but has quite different initial over-segmentation stage and region merging in comparison with other state-of-the-art methods. To obtain high-quality initial partitions for SAR images, we adopt our previously proposed adaptive superpixel generation approach (ALFCE) with linear feature clustering and edge constraint [38]. This method extracts the local gradient ratio pattern from SAR images and proposes to use the feature-ratio-based edge detector with Gauss-shaped window to obtain the edge strength map (ESM). After that, linear feature clustering with edge penalty is implemented to generate superpixels, which are local homogeneous over-segmentation partitions with similar sizes and shapes. Unlike other over-segmentation methods such as watershed transform and TurboPixels, the generated superpixels have high boundary adherence and can well preserve image details. However, there are still some mixed superpixels near the edges, which may contain more than one terrain classes and is heterogeneous, leading to a negative effect on the subsequent region merging. To resolve this issue, we propose a sketch edge map for SAR images and then eliminate mixed superpixels to refine the over-segmentation.

In the region merging stage, this article proposes a rapid superpixel merging for efficient and accurate SAR image segmentation by using the statistical region merging (SRM) framework. The SRM is firstly proposed by Nock and Nielsen [39], which has both statistical and computational advantages. It performs on pixel level and can satisfy the natural image segmentation. Nevertheless, it is rarely applied on SAR images due to the serious speckle. Lang *et al.* [21] proposed to generalize the SRM (GSRM) for polarimetric SAR image segmentation according to the multiplicative property of speckle noise. However, parameters involved in the GSRM are not easy to determine. In addition, the GSRM is performed on pixel levels and not suitable for superpixel merging. To overcome this issue, in this article, we propose to merge superpixels under the SRM framework, leading to fast and accurate SAR image segmentation. The main novelty of this work consists of the following aspects: 1) a new merging order is defined based on the statistical dissimilarity measure and common boundary length penalty, as well as the homogeneity constraint for each superpixel pair. 2) For the merging predicate, we define an adaptive merging threshold according to the image complexity, making the proposed superpixel merging no need to set any merging parameters in advance. 3) More importantly, disjoint set is utilized in this article to map the superpixel pairs to pixel pairs for the sake of fast region merging, which has a low computation cost even with the increasing of superpixel number.

Therefore, in comparison with other state-of-the-art methods, the following characteristics have been made in our proposed segmentation approach.

1) The segmentation is efficient even with the increasing number of initial over-segmentation partitions.
2) Since the processing unit is superpixel, the segmentation is robust to speckle noise.
3) The proposed superpixel merging is free of merging parameters.

Note that the proposed work in this article is quite different from our previously superpixel generation approach, i.e., ALFCE. The ALFCE approach mainly generates superpixels for SAR images, whereas the presented work here merges the generated superpixels to achieve efficient and accurate segmentation for SAR images. Therefore, the applications and methodologies of these two methods are totally different.

II. Initial Superpixel Generation

This section briefly introduces our previously proposed ALFCE with linear feature clustering and edge constraint, namely ALFCE [38] for SAR image superpixel over-segmentation. The feature-based ratio edge detector is described firstly, followed by the superpixel generation algorithm.

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A. Description of Feature-Based Ratio Edge Detector

On the basis of traditional ratio-based edge detectors [28], [29], in [38], a feature-based ratio edge detector is proposed. This method uses the local gradient ratio pattern (LGRP) feature and Gauss-shaped window to achieve SAR image edge detection, leading to an edge map with high precision of localization and low false positive rate.

Inspired by the idea of local binary pattern, LGRP is defined based on the local image intensities of neighbor pixels, as well as the gradient information of SAR images. Therefore, this feature has twofold advantages, which are insensitive to multiplicative speckle noise and robust to small or large image gradient variation in SAR images.

Traditional ratio-based edge detectors usually use two rectangle-shaped subwindows to obtain the ratio of amplitude means as a test statistic for edge detection. However, there easily exist false-edge pixels with these intensity-ratio-based edge detectors, especially in the SAR scenes with low gradient variations. Xiang *et al.* [38] introduced to use the LGRP feature and the Gauss-shaped window instead of the traditional rectangle-shaped window to overcome this problem. Firstly, a Gauss-shaped window is defined, which consists of four parameters, i.e., the window length and width, the spacing between two subwindows, and the angular increment between two orientations. Different from the same weight defined in the rectangle-shaped window, pixels have larger weights if they are close to the center pixel thanks to the Gauss-shaped window, which is good for the edge detection. Then, a set of Gauss-shaped windows with different orientations on each pixel are used to calculate local means of LGRP feature values. Then, the ratio distance of feature values in two subwindows is regarded as the probability of an edge pixel. Finally, the maximum value from different sets of windows constructs the ESM. More detailed explanations are given in [38].

B. Superpixel Generation

ALFCE approach proposed in [38] tries to bridge the weighted local K-means feature clustering and the normalized cut (Ncut) method to achieve SAR image superpixel generation with good boundary adherence and low computation cost. This method adopts a distance metric which measures both the feature similarity and space proximity in the Ncut method. To improve the efficiency, the weighted local K-means is used instead of the traditional eigen-based strategy to minimize the objective function in the original Ncuts method. Therefore, since ALFCE can consider the local clustering of weighted K-means and the global optimization of original Ncuts, the superpixel generation can capture the local and global image properties in the meantime with a low computation cost. It is worth pointing out that in the weighted linear feature clustering, the edge information is taken into consideration to preserve the image structure and further generate semantically meaningful superpixels.

The implementation of ALFCE approach is performed on iteration of the weighted local K-means clustering. The algorithm is summarized as follows [38].

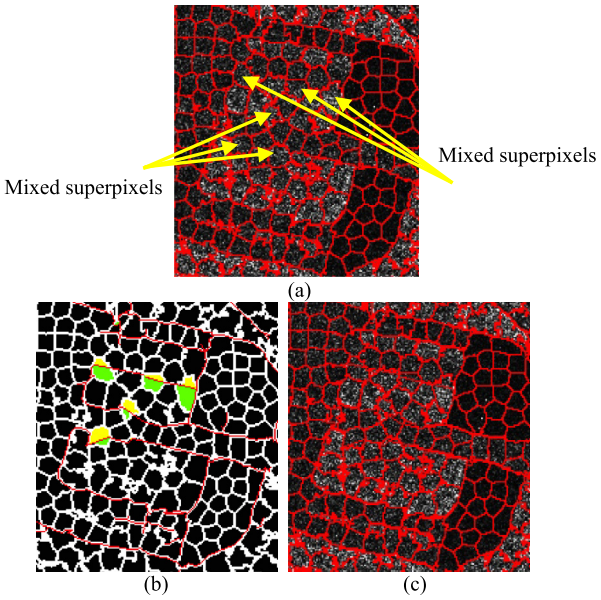


Fig. 1. Mixed superpixels and elimination. (a) Mixed superpixels near edges in an SAR image. (b) Sketch edge splits mixed superpixels. (c) Final superpixels.

1) For each pixel in SAR amplitude image, we calculate the local gradient ratio pattern feature.
2) Edge strength map can be obtained by using the feature-based ratio edge detector.
3) For each pixel pairs, we calculate the Euclidean distance metric of LGRP feature and then approximate it through a positive semidefinite kernel function.
4) We initialize K seeds over the image uniformly at regular grid, where K denotes the number of superpixels. Note that K can be chosen according to the number of looks and scene complexity of SAR images.
5) The weighted local K-means clustering is performed iteratively with a limited search region.
6) Disjointed pixels are eliminated with a postprocessing step.

More information about the ALFCE superpixel generation algorithm and the implementations can be found in [38].

III. Mixed Superpixel Elimination With the Sketch Edge Map

As stated previously, initial over-segmentation superpixels with good boundary adherence can have a positive effect on the region merging. Nevertheless, there usually exist some mixed superpixels near the edges between two homogeneous regions [see Fig. 1(a)]. The reason is that in some superpixel generation algorithms, edge information is not considered. Although the ALFCE approach generates superpixels with edge constraint, the extracted edges are usually discontinuous with missing points due to the speckle noise in SAR images. Therefore, mixed superpixels still exist, which are not homogeneous and have a negative impact on the segmentation performance. In this article, we introduce to use the sketch edge map to eliminate mixed superpixels, which can be regarded as a postprocessing step for the ALFCE approach. Therefore,

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it is efficient for the reason that there is no need to run the ALFCE again.

The sketch edge map contains the structure information of SAR images and the primitive of sketch edge map is effective edges. Some of the edges extracted by the feature-based ratio edge detector are fragmental or discontinuous with missing points, which will lead to mixed superpixels. Therefore, we need to refine the edge map and select effective edges to form a sketch edge map. The procedure is described as follows.

1) We find the endpoints of each edge in the edge map.
2) Each edge is labeled and we can get a list of all the edge points in the current edge.
3) If the edge length, i.e., the total number of pixels on this edge, is less than the minimum length threshold, the edge is regarded a fragmental line and should be rejected.
4) For each endpoint and all other endpoints that are not in the same edge line, we calculate the angles between the two edge lines which the two endpoints belong to. If the angle is smaller than 30°, the two endpoints are selected.
5) The distance between two endpoints is calculated. If the distance is less than the allowable distance threshold, a new edge line is added into the edge map connecting those two endpoints.
6) Repeat step 4 to step 5 for every endpoint.

Note that there are three parameters within the procedure, i.e., the minimum length, the maximum angle, and the distance threshold. The selection of minimum length is related to the number of looks of SAR images. The strength of speckle is strong with a small number of looks. Therefore, the minimum length should be large to avoid the fragmental lines. Generally, we can set the minimum length as five pixels for most of SAR images. Regarding the maximum angle and distance threshold between two endpoints, we generally set them as constant values. In general, if the angle between two endpoints is greater than 30°, or the distance between the two endpoints is greater than ten pixels, the two endpoints cannot be connected because they unlikely belong to the same edge line. It is worth pointing out that the sketch edge map is the optimization of edge detection results. We would rather not connect the endpoints than add too many incorrect edges to the edge map. Therefore, we think that 30° and ten pixels are large enough to connect endpoints which are probably on the same edge line.

After that, for each superpixel, a mixed superpixel check with the sketch edge map is performed. Original superpixels that are crossed by sketch edges are split along the edges into two or more smaller subsuperpixels. Given the superpixel set S_K obtained by ALFCE, and the sketch edge map E_G. The mixed superpixel elimination procedure splits each mixed superpixel P_i into $P_i^{(j)}$ subsuperpixels under the following conditions:

$$\begin{cases} P_i^{(j)} \cap E_G = \varnothing \quad \forall i, j \\ \cup P_i^{(j)} = P_i \\ P_i^{(j)} \cap P_i^{(k)} = \varnothing \quad \forall j \neq k \end{cases} \tag{1}$$

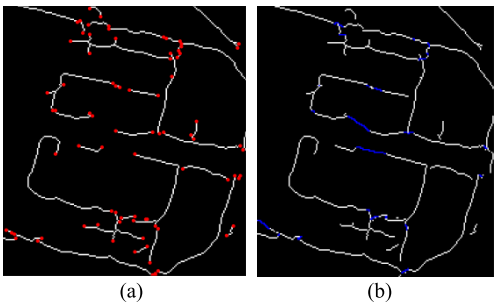


Fig. 2. Illustration of sketch edge map generation. (a) Original edge map with endpoints. (b) Sketch edge map.

where $\cap$ and $\cup$ denote the intersection and union operators and $\varnothing$ is the empty set. The first condition in (1) guarantees that no final superpixels should be crossed by the sketch edge map. The two last conditions define the subsuperpixels $P_i^{(j)}$ as a partition of the parent mixed superpixel P_i.

Fig. 1 depicts the mixed superpixels and the elimination result for an SAR image. It can be seen that after the mixed superpixel check, a sketch edge splits the mixed superpixel into two subsuperpixels and the final superpixel map is better than before. Fig. 2 gives the illustration of sketch edge map generation, where the red points denote the endpoints of each edge and the blue lines are new edges that are helpful to make the discontinuous edge change to be effective.

IV. Fast Superpixel Region Merging Algorithm

A. Traditional SRM Segmentation Approach

The SRM algorithm is actually one of the region growing techniques. The key idea is to start with per pixel and then applying a statistical test on neighboring pixels with an ascending order of gray value differences whether the mean intensities of local regions are sufficiently similar enough to be merged.

There are two key components within this algorithm, i.e., the merging order and merging predicate. The former illustrates which pair of pixels should be merged preferentially among all pair of pixels and the latter is defined as the rule whether two pixels should be merged or not. The merging order can be determined by defining a sort function, which calculates the dissimilarity values of all pixel pairs. Let $|I|$ denotes the number of pixels in an image I. Pixel pairs can be composed of each pixel and its four neighbors. Thus, the image I contains $2|I|$ pairs of adjacent pixels, which can be denoted by S_I. A real-valued gradient function $f(p, p')$ can be utilized as the sort function, which is defined as

$$f(p, p') = |I(p) - I(p')| \tag{2}$$

where p and p' are a pair of adjacent pixels in S_I, $I(p)$, and $I(p')$ are the gray values of two pixels, respectively.

In [39], $\hat{I}$ denotes the perfect theoretical (or ideal) segmentation result of the image I, which exhibits two common homogeneity properties. The first one is that pixels have the same mathematical expectation inside each region and the second is the expectations of adjacent statistical regions

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are different. With this assumption, the merging predicate in [39] is derived as

$$P(R, R') = \begin{cases} \text{true.} & |\bar{R} - \bar{R}'| \leqslant \sqrt{b^2(R) + b^2(R')} \\ \text{false.} & \text{otherwise} \end{cases} \quad (3)$$

where $\bar{R}$ represents the observed average in the region R, and R' denotes the adjacent region of R. $b(R)$ is derived as

$$b(R) = G\sqrt{\frac{1}{2Q|R|} \ln \frac{|R_{|R|}|}{\delta}} \quad (4)$$

where $|R|$ denotes the number of pixels in the region R. $R_{|R|}$ is the set of regions with $|R|$ pixels and $|R_{|R|}| = (|R|+1)^{\min(|R|,G)}$. δ is generally set with a very tiny value like $\delta = 1/(6|I|^2)$. Note that in the best statistical segmentation image $\hat{I}$, the pixel gray $g \in [0, G]$ level and g belongs to one of Q different *r.v.* for the whole image. The parameter Q denotes the statistical complexity of image I [39]. For instance, if the image contains rich information such as many heterogeneous regions and complex scenes, Q should be high, leading to more segments in the final segmentation map.

From the above, it can be seen that the traditional SRM algorithm has some limitations for superpixel merging in SAR images. First of all, the SRM starts from pixels and the merging order is determined by the gradient information of adjacent pixels, which would be sensitive to multiplicative speckle in SAR images. In addition, it cannot be directly applied on superpixel merging due to its pixel-based strategy. Second, the statistical information of SAR images is complex, making the image statistical complexity parameter Q not easy to determine. Therefore, how to define the new merging order and merging predicate of superpixels should be further investigated.

B. Merging Order for Superpixel Pairs With Multiple Cues

The merging order of adjacent superpixel pairs has a great influence on the quality of final segmentation results. Superpixels with a smaller dissimilarity should be merged prior to those with a larger one. In this article, we present a dissimilarity measure for superpixel pairs, which is based on the following rules.

1) Since superpixels are groups of homogeneous pixels, the dissimilarity measure needs to consider the statistical SAR image intensity/amplitude information between two superpixels, which should be robust to speckle noise.
2) The spatial relationship between superpixels should be considered. For instance, adjacent superpixels with a longer common length should have a greater chance to be merged. Superpixels with smaller sizes will be more likely merged than those with larger sizes, which can eliminate the fragmentized segments with small sizes. Note that the size of a superpixel is the number of pixels in it.
3) Superpixels can be hardly merged if there exist image edges between them.
4) A smaller homogeneity difference will result in likely merging between two superpixels, which should be considered in the dissimilarity measure.

For SAR images, superpixels with similar backscattering tend to exhibit close statistical intensity or amplitude values. However, due to the multiplicative speckle noise, traditional dissimilarity measurements of intensity/amplitude such as the Euclidean distance are not applicable to SAR images [40]. To measure the dissimilarity of two pixels, Feng *et al.* [40] proposed a ratio distance for SAR image speckle reduction. Yuan *et al.* [41] utilized a pixel ratio measurement for SAR target recognition, which is derived from the ratio of intensity average in a local neighbor window. Apart from the ratio distance, Guan *et al.* [42] proposed to use the generalized likelihood ratio similarity to measure the intensity similarity between SAR noisy pixels. Note that these measurements are designed for pixels not superpixels; Yu *et al.* [20] proposed to extract brightness and texture features from two regions and then measure the feature distance. Zhang *et al.* [3] utilized the Kuiper's distance between two histograms estimated from two regions as the measurement. However, the computations are not efficient. Here, we use the sample mean amplitudes of adjacent superpixels to derive the backscattering difference. The sample means of two superpixels R and R' are computed by

$$\bar{m}_1 = \frac{1}{|R|}\sum_{p\in R} I(p) \quad \bar{m}_2 = \frac{1}{|R'|}\sum_{p\in R'} I(p). \quad (5)$$

Assuming the two superpixels have constant backscattering m_1 and m_2, then based on the multiplicative speckle model of SAR images, the errors of the estimates can be written by

$$\begin{aligned} \bar{m}_1 - m_1 &= \frac{m_1}{|R|}\sum_{p\in R}(\varepsilon(p) - 1) \\ \bar{m}_2 - m_2 &= \frac{m_2}{|R'|}\sum_{p\in R'}(\varepsilon(p) - 1) \end{aligned} \quad (6)$$

where the speckle $\varepsilon(p)$ in the L-look SAR amplitude image has unit mean and variance $(4-\pi)/(L\pi)$ [28]; then, we have the variance of $\bar{m}_1 = \bar{m}_2$ as

$$\text{var}\{(\bar{m}_1 - \bar{m}_2) - (m_1 - m_2)\} = \frac{(4-\pi)}{L\pi}\left(\frac{m_1^2}{|R|} + \frac{m_2^2}{|R'|}\right). \quad (7)$$

The backscattering difference between two superpixels can be derived as [15]

$$\sigma_0(R, R') = \frac{\sqrt{L\pi}\,|\bar{m}_1 - \bar{m}_2|}{\sqrt{4-\pi}\sqrt{\frac{m_1^2}{|R|} + \frac{m_2^2}{|R'|}}}. \quad (8)$$

For the exchangeable property, i.e., $\sigma_0(R, R') = \sigma_0(R', R)$, and the backscattering reflectances m_1 and m_2 can be replaced by their estimates [sample means in (5)], the backscattering

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dissimilarity is rewritten as

$$\sigma_0(R, R') = \frac{\sqrt{L\pi}\left(1 - \min\left\{\frac{\bar{m}_1}{\bar{m}_2}, \frac{\bar{m}_2}{\bar{m}_1}\right\}\right)}{\sqrt{4-\pi}\sqrt{\frac{1}{|R|}\left(\min\left\{\frac{\bar{m}_1}{\bar{m}_2}, 1\right\}\right)^2 + \frac{1}{|R'|}\left(\min\left\{\frac{\bar{m}_2}{\bar{m}_1}, 1\right\}\right)^2}}. \tag{9}$$

It can be seen that $\sigma_0(R, R')$ is a function of the ratio of average amplitude of two superpixels, which would be robust to speckle. Furthermore, it is nonnegative, exchangeable, and can be simply computed.

The spatial relationship between superpixels and the image edges have a great influence on the merging. In this article, we consider the sizes and common boundary of superpixel pairs, as well as the ESM to derive a spatial dissimilarity between two superpixels. It is depicted as

$$\sigma_\zeta(R, R') = \frac{\min(|R|, |R'|) \cdot E_\varepsilon}{\mathrm{Len}(\mathscr{B}(R) \cap \mathscr{B}(R'))} \tag{10}$$

where $\mathscr{B}(R)$ denotes the boundary pixels of superpixel R and the denominator is the length of common boundary of two superpixels. E_ε depicts the average edge strength of the common boundary pixels. Different from the boundary length penalty term in [2] and [15], the proposed spatial dissimilarity not only considers the superpixel sizes and the common boundary length, but also considers the edge strength of common boundary pixels, which can prevent the merging of superpixels with strong edge strength.

Generally, superpixels are local homogeneous. The smaller the homogeneity dissimilarity of two superpixels, the higher the possibility of being merged. Here, we define the homogeneity dissimilarity as

$$\sigma_{\mathscr{H}}(R, R') = |\mathscr{H}(R) - \mathscr{H}(R')| \tag{11}$$

where $\mathscr{H}(R)$ denotes the homogeneity of superpixel R and is defined as

$$\mathscr{H}(R) = \frac{1}{|R|}\sum_{p \in R}\mathscr{H}(p), \quad \mathscr{H} = \frac{\mathrm{ENL}}{\mathrm{ESM}}. \tag{12}$$

$\mathscr{H}$ is the homogeneity map of SAR images and can be defined as the ratio of estimated equivalent number of looks (ENL) and the ESM [43].

Finally, we combine the backscattering, spatial, and homogeneity dissimilarities to achieve the final dissimilarity measure of two superpixels R and R' as

$$\sigma(R, R') = \sigma_0(R, R') + \sigma_{\mathcal{G}}(R, R') + \sigma_{\mathscr{H}}(R, R'). \tag{13}$$

C. Adaptive Merging Predicate

First of all, we review the theory based on which the merging predicate of SRM is derived.

Theorem 1: Let $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ be a family of n independent *r.v.* with X_k taking values in a set A_k for each k. A real-valued function f is defined on $\Pi_k A_k$ which satisfies $|f(\mathbf{x}) - f(\mathbf{x}')| \leqslant c_k$ whenever the vectors $\mathbf{x}$ and $\mathbf{x}'$ differ only in the kth coordinate. Suppose that μ is the expected value of $f(\mathbf{X})$. Then, for any $\tau \geqslant 0$

$$\Pr(f(\mathbf{X}) - \mu \geqslant \tau) \leqslant \exp\left(-2\tau^2 \Big/ \sum_k (c_k)^2\right). \tag{14}$$

With *Theorem 1*, *Corollary 2* can be deduced as follows [39].

Corollary 2: Let (R, R') denote a pair of regions of the image I. $\bar{R}$ is the average gray value of region R. $\forall 0 < \delta < 1$, we have

$$\Pr\left(|(\bar{R} - \bar{R}') - E(\bar{R} - \bar{R}')| \geqslant G\sqrt{\frac{1}{2Q}\left(\frac{1}{|R|} + \frac{1}{|R'|}\right)\ln\frac{2}{\delta}}\right) \leqslant \delta \tag{15}$$

where the proof can be referred to [39]. As we discussed before, the original merging predicate in SRM is based on the average gray value difference of two regions, which cannot be applied on SAR images. In this article, we present a new adaptive merging predicate for SAR images, making the merging threshold be adaptive to SAR image content and free of parameters.

Corollary 3: For each superpixel pair (R, R'), we denote their symmetric dissimilarity measure as $\sigma(R, R')$. $\forall 0 < \delta < 1$, the probability no more than can be derived as

$$|\sigma(R, R') - E(\sigma(R, R'))| \geqslant \sqrt{\frac{(\max[\sigma(R, R')])^2}{2Q}\left(\frac{1}{|R|} + \frac{1}{|R'|}\right)\ln\frac{2}{\delta}}. \tag{16}$$

Proof: We consider to shift the value of outcome of one *r.v.* among the $Q(|R| + |R'|)$ possible for the superpixel pair (R, R'). Thus, $\sigma(R, R')$ is generally subject to a variation of at most $c_R = \max[\sigma(R, R')]/(Q|R|)$ (among $Q|R|$ possible) when performing on the superpixel R and at most $c_{R'} = \max[\sigma(R, R')]/(Q|R'|)$ for a change inside superpixel R'. Then, we get

$$\sum_k (c_k)^2 = Q(|R|(c_R)^2 + |R'|(c_{R'})^2) = \frac{(\max[\sigma(R, R')])^2}{Q}\left(\frac{1}{|R|} + \frac{1}{|R'|}\right). \tag{17}$$

The inequality (14) can be rewritten as

$$\Pr(f(\mathbf{X}) - \mu \geqslant b_\sigma(\mathbf{X})) \leqslant \delta \tag{18}$$

where $b_\sigma(\mathbf{X})$ is the merging threshold and denoted by

$$b_\sigma(R, R') = \sqrt{\frac{(\max[\sigma(R, R')])^2}{2Q}\left(\frac{1}{|R|} + \frac{1}{|R'|}\right)\ln\frac{2}{\delta}} \tag{19}$$

for the superpixel pair (R, R'). Then, the *Corollary 3* can be obtained.

Note that the maximum of $\sigma_0(R, R')$, $\sigma_{\mathcal{G}}(R, R')$, and $\sigma_{\mathscr{H}}(R, R')$ can be derived as $\sqrt{L\pi}\sqrt{\max(|R|, |R'|)}/\sqrt{4-\pi}$, $\min(|R|, |R|)$ and 1, respectively, for the reason that R_ε and $\mathscr{H}$ will be normalized to $[0, 1]$ in the implementation. Therefore, $\max[\sigma(R, R')]$ in (19) can be denoted by $\sqrt{L\pi}\sqrt{\max(|R|, |R'|)}/\sqrt{4-\pi} + \min(|R|, |R'|) + 1$. δ can be set as $1/(6|I|^2)$. The factor Q exhibits similar function like the

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homogeneity map $\mathscr{H}$ since they both measure the complexity of SAR scene structure. Thus, the merging threshold for (R, R') can be rewritten as

$$b_\sigma(R, R') = \sqrt{\frac{(\max[\sigma(R, R')])^2}{2(1-\mathscr{H})}\left(\frac{1}{|R|}+\frac{1}{|R'|}\right)\ln\frac{2}{\delta}}. \quad (20)$$

Finally, the new adaptive merging predicate for superpixels in SAR images can be obtained as

$$P(R, R') = \begin{cases} \text{true.} & |\sigma(R, R') - E(\sigma(R, R'))| \leqslant b_\sigma(R, R') \\ \text{false.} & \text{otherwise.} \end{cases} \quad (21)$$

D. Discussion on the Computational Complexity and the Fast Algorithm Implementation

Considering that this article mainly pays attention to the superpixel merging, here we only give the discussion on the computational complexity analysis of the merging stage. In the proposed approach, the computation cost consists of the merging order and merging predicate. In the calculation of dissimilarity measures of superpixel pairs, the time complexity can be denoted by $O(K \cdot \Omega)$, where K is the number of superpixels for each superpixel and Ω represents the number of adjacent superpixels. After the calculation of dissimilarity for each superpixel pair, the merging order based on radix sorting with $\sigma(R, R')$ as the key can be obtained. Note that we do not update the list of merging order after merging each superpixel pair, and the time complexity will remain the same during the whole merging stage. The merging stage afterward is performed on the basis of each superpixel pair by comparing the values of $\sigma(R, R')$ and $b_\sigma(R, R')$, which also yields a time complexity as $O(K \cdot \Omega)$. Therefore, the whole time complexity can be denoted by $2O(K \cdot \Omega)$, which is quite related to the superpixel number and the neighborhood in SAR images.

On the basis of aforementioned analysis, it is worth pointing out that with the increasing of superpixel number, the time complexity of our proposed merging approach will increase dramatically, resulting in an inefficient SAR image segmentation. In addition, the number of adjacent superpixels for each superpixel is not easy to determine due to irregular shape and size of the superpixel. To resolve these problems, we propose to use the disjoint set in this article to map the superpixel pairs to pixel pairs for the sake of fast merging, which has a low computation cost even with the increasing of superpixel number.

The disjoint set is a data structure that tracks a set of elements partitioned into several disjoint (nonoverlapping) subsets. It provides near-constant-time operations like adding new sets (*MakeSet* operation), merging existing sets (*Union* operation), and determining whether elements are in the same set (*Find* operation). The *MakeSet* operation makes a new set with a unique number, a rank of zero, and a parent pointer to itself. The *Find* operation follows the chain of parent pointers from the current element up the tree until it reaches a root element, whose parent is itself. The *Union* operation first uses the *Find* to determine the roots of two trees. Then, the two trees are merged by attaching the root of one to another if the roots are distinct. Fig. 3 gives the diagram of disjoint-set operations.

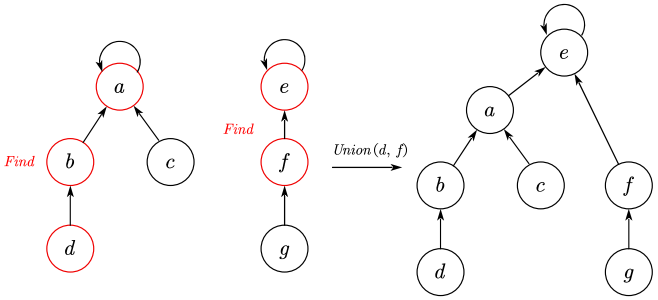


Fig. 3. Schematic of the disjoint-set operations, where the elements with red circles indicate the results of *Find* operation.

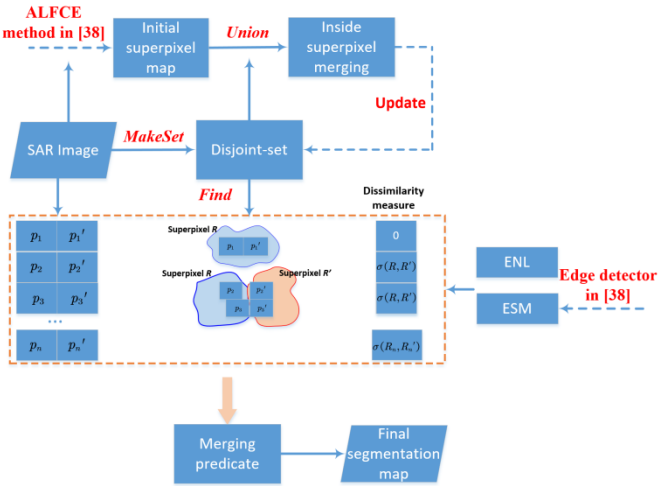


Fig. 4. Flowchart of the proposed algorithm.

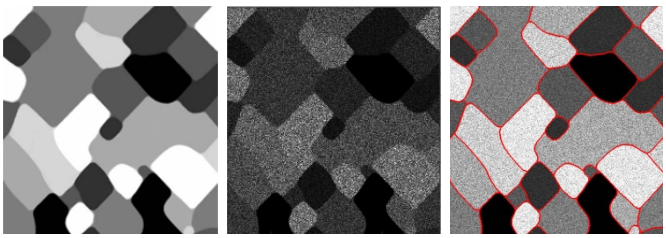
Fig. 4 and Table I give the flowchart and implementation details of our proposed superpixel merging algorithm. It can be seen from Table I that step 3 is the calculation of merging order list for all pair of pixels, which consists of five sub steps. Equations (5)–(13) should be calculated for each pixel pair. Therefore, the computation cost of our proposed superpixel merging approach mainly comes from the determination of merging order list for all the pixel pairs. Even though there are more pixel pairs than superpixel pairs, dissimilarity measures between most of the pixel pairs do not need to be calculated, i.e., zero, except for pixel pairs at the boundary of adjacent superpixels. Due to the constraint of superpixels, the dissimilarity measure between pixel pairs at the superpixel boundaries should be calculated only once. Then, all the pixel pairs which belong to the same superpixel pair have the same dissimilarity. Similarly, in the merging stage, if the superpixel pair is merged, all the pixel pairs in it should be merged automatically. With the highly efficient disjoint-set data structure, the superpixel pairs are mapped into pixel pairs and the time costs of proposed merging method will not change significantly even with the increasing of superpixel number. Therefore, our proposed method is efficient, which will be demonstrated in the following experiments with synthetic and real SAR data sets.

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TABLE I
FAST SUPERPIXEL MERGING ALGORITHM FOR SAR IMAGES

Input: The edge strength map ESM, the SAR image I and the superpixel label for each pixel.
Output: Superpixel merging result and final segmentation map.

1. For each pixel in the SAR image I, calculate the equivalent number of looks and then initialize the disjoint-set with size, rank and parent by using the *MakeSet* operation.
2. Inside each superpixel, perform the *Union* operation for the pixels and then update the disjoint-set.
3. Map the superpixel pairs to pixel pairs for the sake of fast merging. On the basis of superpixel constraint, the merging order list is achieved based on pixel pairs.
   a) Each pixel with its eight neighbor pixels form adjacent pixel pairs. Thus the image I totally has $4|I|$ adjacent pixel pairs.
   b) For each pixel pair, the dissimilarity measure should be zero if they belong to the same superpixel. Otherwise, perform *Find* operation to obtain their parent pointers.
   c) Based on the parent pointers we can get two superpixels which the pixel pair belongs to. If the dissimilarity measure of these two superpixels is not calculated, the dissimilarity measure can be obtained by using (13), and then recorded. Otherwise, if the dissimilarity measure of the two superpixels can be found in the record, indicating that the dissimilarity measure has already been calculated and there is no need to calculate it again.
   d) Perform *b)* and *c)* for all the pixel pairs.
   e) Sort the pixel pairs in increasing order of the dissimilarity measure and obtain the merging order list.
4. For each pixel pair with the dissimilarity measure not zero in the merging list, perform *Find* operation to obtain the superpixel pair they belong to, then use *Union* operation to merge the two superpixels if the merging predicate returns true.
5. Based on the final disjoint-set tree of image I, the final superpixel merging result and segmentation map can be obtained.



(a) (b) (c)

Fig. 5. Simulated SAR image with carton texture and the ground truth map. (a) Carton texture image. (b) Four-look simulated SAR image. (c) Ground truth segmentation map.

V. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, the results of our proposed algorithm and three comparative methods are reported for synthetic and real SAR images. First of all, the SAR image data sets and three comparative methods are described, followed by the performance evaluation and comparison of different methods. Several quantitative indexes, such as the precision-recall (P-R) framework, the segment covering criteria (SCC), and the variation of information (VI), are adopted for evaluation. Finally, we give the computational loads of different methods.

A. Descriptions of Data Sets and Comparative Methods

Fig. 5(a) shows the carton texture image and Fig. 5(b) shows the corresponding simulated SAR image with four-look Gamma speckle. The image size is 512×479. Fig. 5(c) gives the ground truth segmentation map. Fig. 6 presents four amplitude images acquired by different airborne and spaceborne SAR sensors. Fig. 6(a) shows a four-look X-band image from the CETC airborne SAR sensor of China, where the resolution is 0.5 m and the image size is 587×690. Fig. 6(b) shows a single-look MiniSAR Ka-band airborne SAR image, which was acquired by the Sandia National Laboratories of USA. The image resolution is 0.1 m and the image size is 7.6×780. Fig. 6(c) shows a spaceborne three-look X-band image from the ICEYE SAR satellite of Finland, where the resolution is 0.55 m and the image size is 890×1025. Fig. 6(d) shows a two-look spaceborne C-band image acquired from the Chinese Gaofen-3 SAR satellite, where the resolution is 1 m and the image size is 887×1016. It can be seen from Fig. 6 that the four SAR image scenes contain roads, built-up areas, vehicles, bare soils, vegetated areas, and so on, which are complex and suitable for the demonstration of image segmentation task. Fig. 7 gives the corresponding ground truth maps of four data sets, which are obtained manually.

In this article, we choose three region-based SAR image segmentation methods for performance comparison. The first one is hierarchical region merging method with gradually increasing orientated edge strength penalty (GIOESP) [3]. The second one is fast region merging approach with RCBLP [15]. The last one is a newly published SAR image segmentation method, i.e., the region-merging method with texture pattern attention (RTPA) [18]. These three methods are all based on the initial over-segmentation results and focus on the efficient region merging strategy to achieve final segmentation for SAR images. It should be noted that the parameter settings of comparison methods are set as optimal values according to independently experiments with 10 times for each SAR image. In terms of the number of initial over-segmentation partitions (superpixel number), all the methods choose the same value for fair comparison. Note that all the methods are implemented using the MATLAB language on a PC with 2.2-GHz i7 CPU.

B. Experimental Results of Different Methods for the Synthetic and Real SAR Data Sets

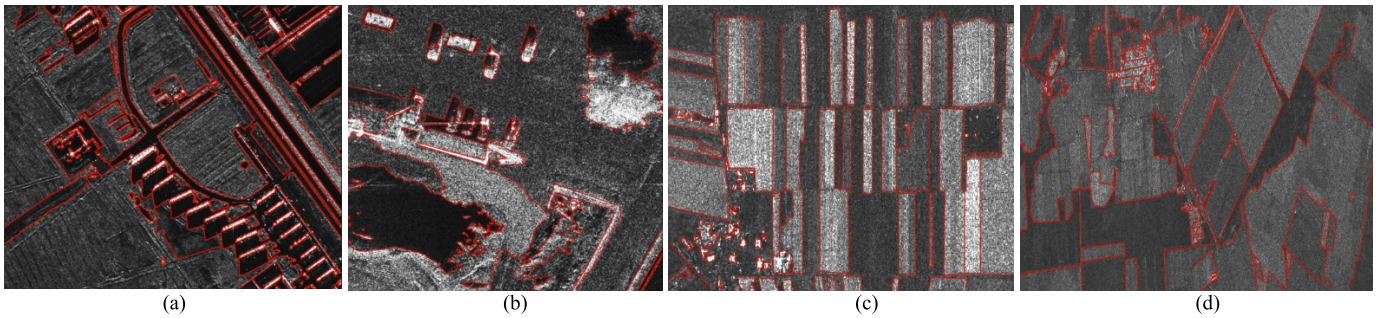
Fig. 8 presents the segmentation results of different methods for the synthetic SAR image, where the first column shows the segmentation boundaries of final segments and the right column denotes the corresponding average segmentation images. The number of initial over-segmentation partitions for the four methods is set to 1500. Fig. 8(a) and (b) shows the results of GIOESP method, where we can see that most

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(a) (b) (c) (d)

Fig. 6. The amplitude images of four SAR data sets. (a) CETC X-band data. (b) MiniSAR Ka-band data. (c) ICEYE X-band data. (d) Gaofen-3 C-band data.



(a) (b) (c) (d)

Fig. 7. Ground truths of the four SAR images, which are generated manually.

of the regions can be correctly discriminated; nevertheless, there are some over-segmentation errors, such as the small regions near the boundaries. The results in Fig. 8(c) and (d) demonstrate that the RCBLP method can effectively reduce the over-segmentation errors. In addition, the segmentation boundaries are smooth and continuous, indicating that different regions can be well discriminated. Nevertheless, there are some over-merging errors, such as the areas marked with yellow rectangle. The results of RTPA depicted in Fig. 8(e) and (f) seem better than the former two methods, where most of the regions can be well discriminated with less over-merging and over-segmentation errors. Fig. 8(g) and (h) shows the results of our proposed method, where we can find that it performs the best among different methods. Almost all of the regions can be well discriminated, including the areas marked with yellow rectangles.

Fig. 9 gives the segmentation results of different methods for the CETC X-band data, where the top row denotes segmentation boundaries of the final segments and the second row shows the corresponding average segmentation images of different methods, respectively. The number of initial over-segmentation partitions for the four methods is set to 1500. From Fig. 9(a), we can find that using the GIOESP, there are many over-segmentation regions in the result although most of the regions can be correctly discriminated. For example, the homogeneous farm area is over-segmented into many regions once there is a little bit gray value difference, indicating that the GIOESP method takes more consideration of the image amplitude information and less consideration of the image texture or homogeneity. Furthermore, the man-made targets such as the buildings and small point targets are not well preserved, such as the area marked with yellow circle. The result of RCBLP in Fig. 9(b) seems better than the GIOESP with less over-segmentation and better preservation of man-made targets. Nevertheless, there are still some over-segmentation results within the river area. Fig. 9(c) gives the result of RTPA method, where we can see that most of the regions can be correctly discriminated with less over-segmentation results than Fig. 9(a) and (b), especially in the natural areas. However, in the built-up areas, the segmentation result is not very satisfactory, where there exist some over-merging results, leading to details missing of the buildings. Some small-size buildings cannot be well segmented. The segmentation result of our proposed approach is given in Fig. 9(d), where it can be seen that the built-up areas and natural areas are all correctly segmented. The over-segmentation errors are less than other methods and the buildings as well as some point targets can be well preserved, as shown in the selected areas. The reason is that our proposed method not only considers the amplitude information of SAR images, but also utilizes the spatial relationship and homogeneity information of superpixels in the definition of merging cost. More importantly, we incorporate the SAR image homogeneity into the merging predicate, resulting in an adaptive superpixel merging with less over-segmentation errors than other methods. Therefore, it can be demonstrated that our method outperforms other three approaches in terms of over-segmentation errors and the preservation ability of SAR image details.

In order to compare the segmentation results of different methods clearly, we select one area in Fig. 9 and display the enlarged results in Fig. 10. Fig. 10(a) presents the corresponding original SAR image of the select area

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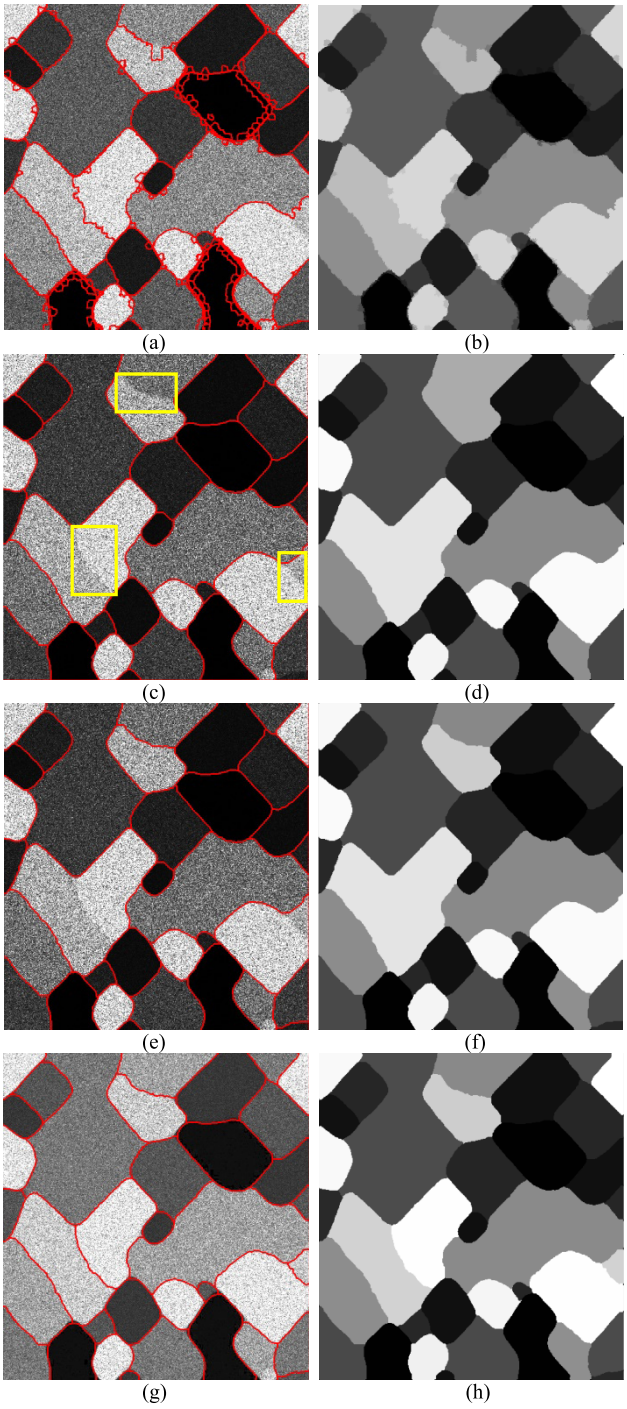


Fig. 8. Segmentation results of different methods for the synthetic SAR data. (a) and (b) GIOESP. (c) and (d) RCBLP. (e) and (f) RTPA(g) and (h) Proposed method.

and Fig. 10(e) shows the segmentation result of the proposed method. Fig. 10(b)–(d) denotes the homogeneity map, superpixels, and segmentation result of the proposed method. The average segmentation images of the proposed method, the GIOESP, RCBLP, and RTPA are given in Fig. 10(f)–(h), respectively. It can be seen that there are buildings, vegetated areas, roads, and isolated small man-made targets in this selected area. Although the GIOESP, RCBLP, and RTPA can correctly segment the vegetated areas and roads, the buildings and some point targets are not well segmented. In contrast, the proposed method can achieve fine segmentation with good preservation of image details, as shown in Fig. 10(d) and (e), where the buildings and point targets can be clearly discriminated. Different land covers can be well segmented with clear boundaries. Therefore, the proposed method has advantages in high-resolution SAR image segmentation.

Fig. 11 gives the segmentation results of different methods for the MiniSAR Ka-band data, where the backscattering is complex due to the high frequency of Ka-band. The number of initial over-segmentation partitions for the four methods is set to 2000. Fig. 11(a) and (e) shows the result of GIOESP method, where it can be found that there are some over-segmentation errors. Some homogeneous areas such as the shadow areas of trees and cars are oversegmented into several regions. The result of RCBLP in Fig. 11(b) and (f) is worse than the GIOESP. The over-segmentation errors within the homogeneous areas are common; furthermore, some small targets such as cars are missing in the segmentation result. Fig. 11(c) and (g) shows the results of RTPA, which seems better than the GIOESP and RCBLP. There are few over-segmentation errors. Although the results within natural areas are satisfactory, there exist some over-merging errors. Some small man-made targets cannot be well preserved. The last column denotes the result of proposed method, where we can see that it shows good segmentation ability in both the homogeneous and heterogeneous areas thanks to the adaptive superpixel merging.

Fig. 12 shows the final segmentation results for the ICEYE X-band SAR image. The number of initial over-segmentation partitions is set to 2500 for the four methods. Similar to the results depicted in Fig. 11, the results of GIOESP seem to be fine in most of the regions. However, there are some over-segmentation errors if we try to preserve the small buildings. In addition, the segmentation boundaries between different regions cannot well adhere to the real image boundaries. Compared with Fig. 12(a), the results of RCBLP and RTPA are worse in terms of the over-merging errors. Some small man-made structures cannot be well preserved and some regions with boundaries are merged, leading to an incorrect segmentation. Nevertheless, it should be noted that the segmentation boundaries of these two methods have more correct locations than GIOESP, especially the RTPA method. The reason is that this method considers more information of the texture, leading to more accurate segmentation result in natural areas in comparison with the GIOESP. Fig. 12(d) depicts the segmentation results of proposed approach, where we can find that the regions can be clearly discriminated with correct segmentation boundaries. Furthermore, the small man-made structures can also be well preserved. The reason is that the proposed method considers the amplitude, edge, and homogeneity information in the merging cost, making the segmentation boundaries correct. Moreover, the adaptive superpixel merging predicate makes it possible to effectively preserve small point targets while effectively segmenting other homogeneous areas.

Fig. 13 depicts the segmentation results of different methods for the GF-3 C-band data. The number of initial

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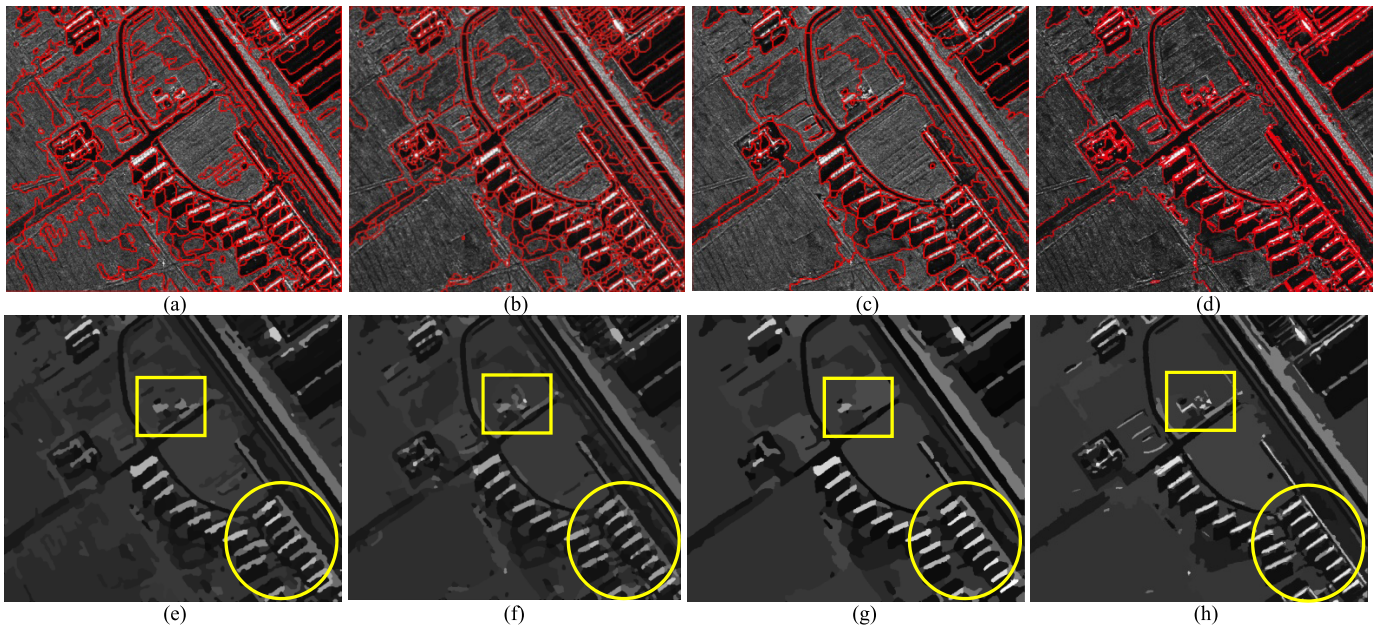


Fig. 9. Segmentation results of different methods for the CETC X-band data. (a) GIOESP. (b) RCBLP. (c) RTPA. (d) The proposed method. (e)–(h) Corresponding average segmentation images of different methods, respectively.

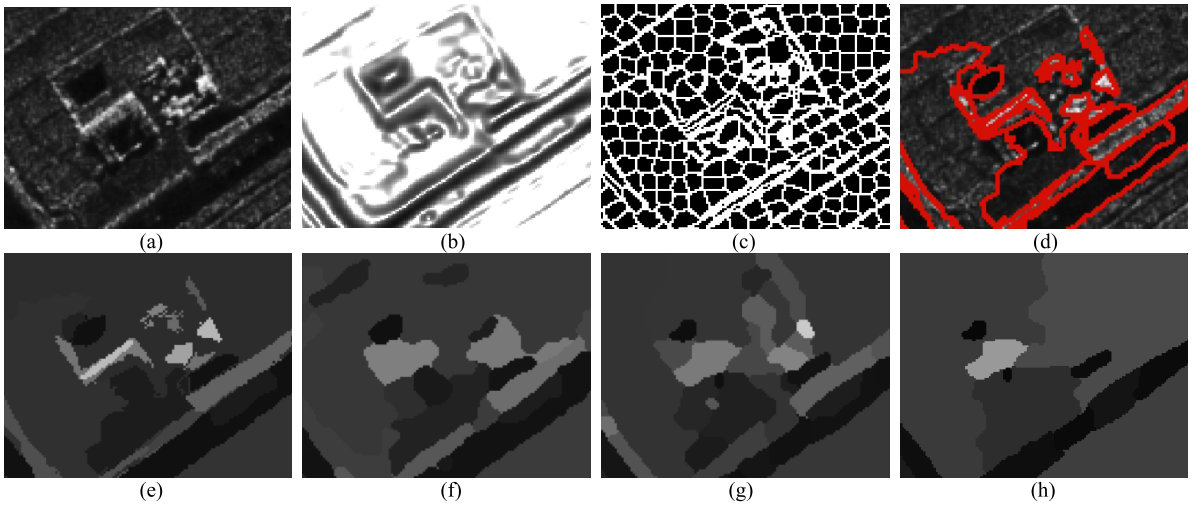


Fig. 10. Detailed comparison of different methods for the select area in Fig. 9. (a) Original SAR image. (b)–(d) Homogeneity map, superpixels, and segmentation result of the proposed method for the selected area. (e)–(h) Average segmentation images of the proposed method, the GIOESP, RCBLP, and RTPA, respectively.

over-segmentation partitions is set to 2500. It can be seen that the test SAR data have a low image contrast. The region boundaries between different land covers are not very clear. From Fig. 13(a), it can be found that the GIOESP can well segment the regions within most of the scenes, indicating that the region merging with orientated edge strength weighted Kuiper's distance is effective to the low image contrast. Furthermore, the man-made structures can be discriminated, even the small isolated buildings, such as the area marked with yellow rectangle in Fig. 13(e). Nevertheless, there are many over-segmentation errors in the homogeneous areas. The results of RCBLP and RTPA in Fig. 13(b) and (c) have few over-segmentation errors but many image details absence. The built-up areas cannot well be extracted and preserved, especially the small buildings. Fig. 13(d) gives the final results of the proposed method, where we can see that the built-up areas with structure details can be clearly discriminated and preserved.

Moreover, most of the natural homogeneous areas can be well segmented although there are few over-segmentation errors due to the low edge strength. In comparison with other three methods, it can be seen that the proposed method has the best ability which balances the over-segmentation errors and preservation of the image structure details, leading to a high segmentation accuracy.

C. Quantitative Evaluations of Different Methods

For the sake of quantitative evaluation of different methods for SAR image segmentation, in this article, we adopt some indices used in optical image segmentation, which are the P-R framework, the SCC, and the VI. The P-R framework is derived based on the comparison between segmentation boundaries and the ground truth, where the precision denotes the fraction of boundaries those are true from the overall detected boundaries and the recall is defined as the probability

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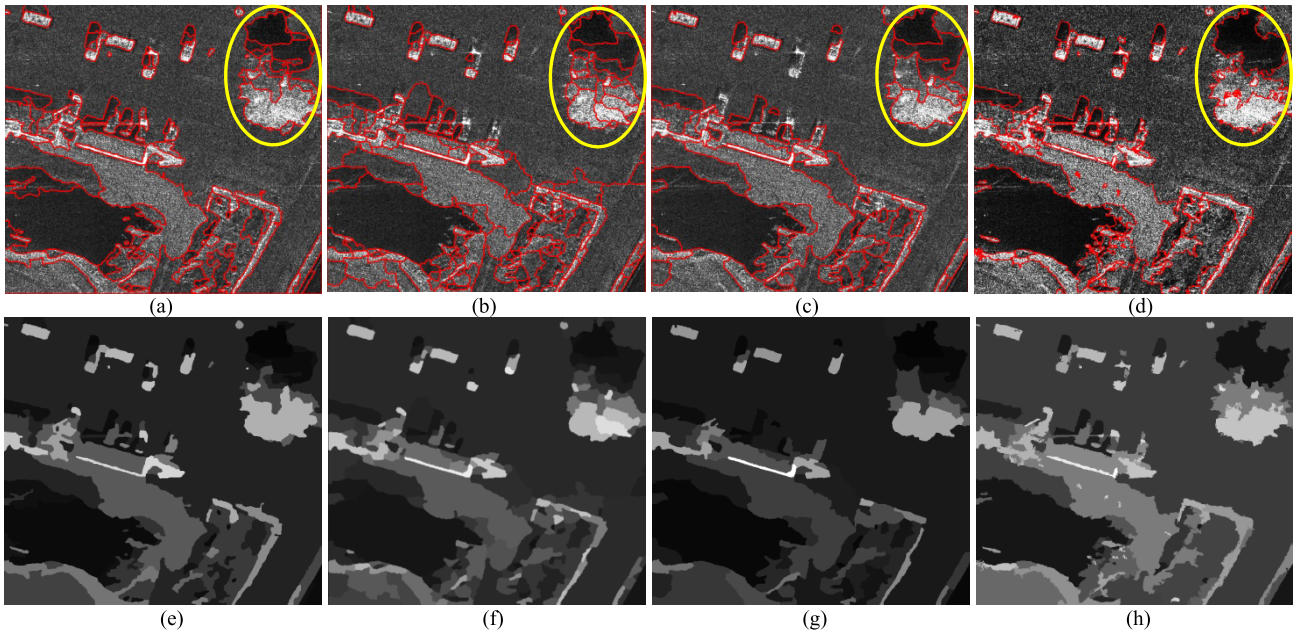


Fig. 11. Segmentation results of different methods for the MiniSAR Ka band data. (a) GIOESP. (b) RCBLP. (c) RTPA. (d) Proposed method. (e)–(h) Corresponding average segmentation images of different methods, respectively.

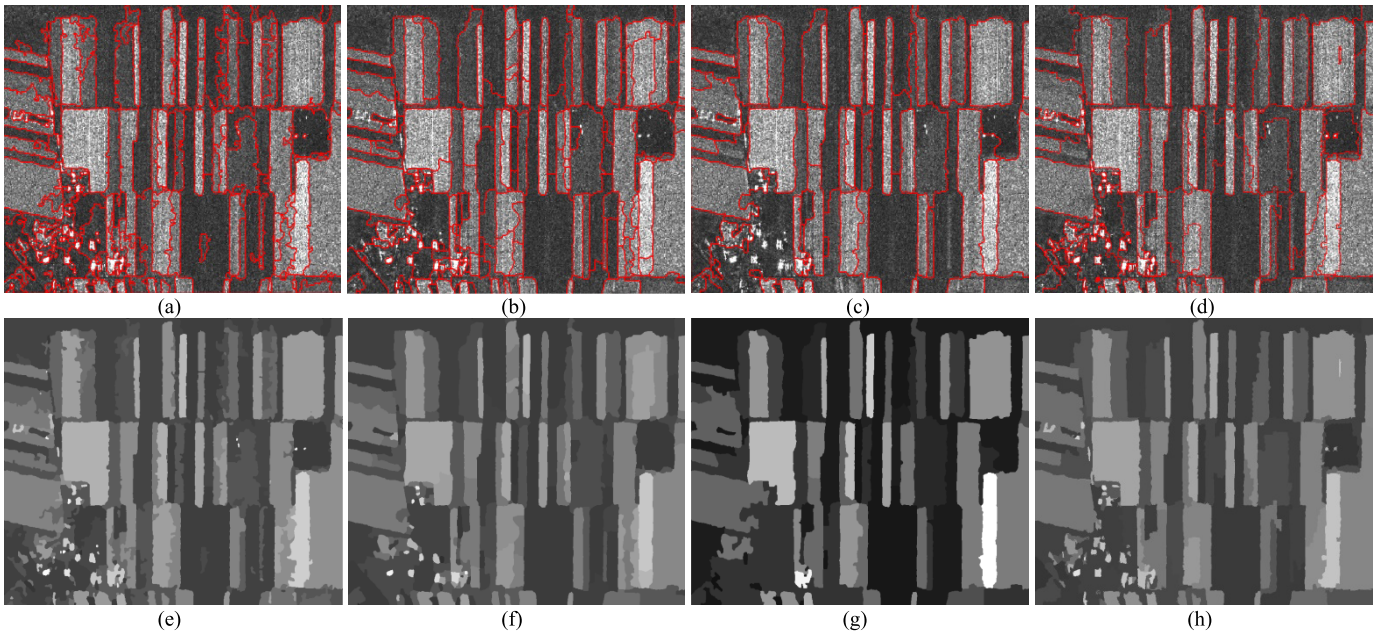


Fig. 12. Segmentation results of different methods for the ICEYE X-band data. (a) GIOESP. (b) RCBLP. (c) RTPA. (d) Proposed method. (e)–(h) Corresponding average segmentation images of different methods, respectively.

of the edge pixels in the overall ground-truth boundaries to be detected. Let $\mathrm{GT}(x)$ be the binary ground truth map and $\mathrm{Seg}(x)$ be the binary segmentation boundary map. Then, the precision (Pre) and recall (Rec) can be defined as [44]

$$\mathrm{Pre} = \frac{\sum \mathrm{GT}(x) \cdot \mathrm{Seg}(x)}{\sum \mathrm{Seg}(x)}, \quad \mathrm{Rec} = \frac{\sum \mathrm{GT}(x) \cdot \mathrm{Seg}(x)}{\sum \mathrm{GT}(x)}. \tag{22}$$

Furthermore, these two measures can be combined together as a single F-measure, which gives an overall boundary quality assessment for the segmentation method. A larger F-measure indicates that the segmentation boundary is closer to the ground truth. The calculation can be represented as

$$F = \frac{2 \cdot \mathrm{Pre} \cdot \mathrm{Rec}}{\mathrm{Pre} + \mathrm{Rec}}. \tag{23}$$

Note that the P-R curves are usually used for the evaluation, which can be obtained by changing the segmentation parameters of a method. In this article, we change the number of initial over-segmentation partitions (superpixels) and then perform region merging to obtain different segmentation results.

In addition to the assessment of segmentation boundary quality, there are some region quality evaluation measures such as the SCC and the VI [45]. Let $\Theta = \{R_m, m = 1, 2, \ldots, M\}$

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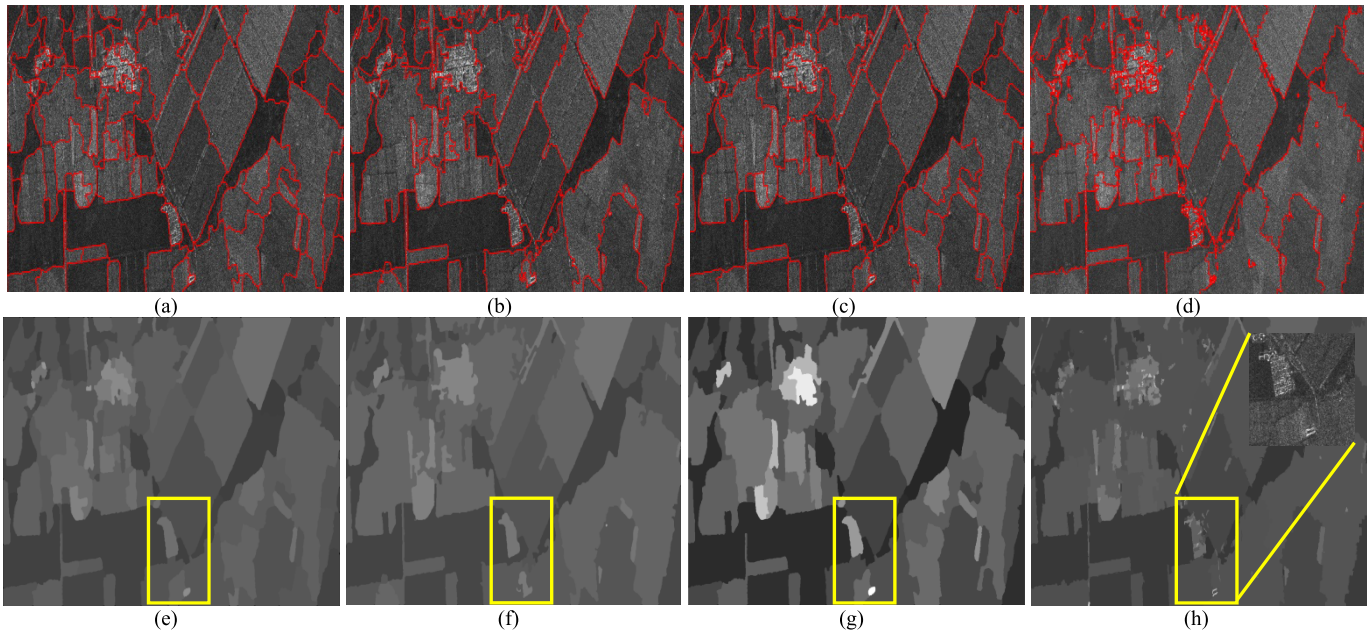


Fig. 13. Segmentation results of different methods for the GF-3 C-band data. (a) GIOESP. (b) RCBLP. (c) RTPA. (d) Proposed method. (e)–(h) Corresponding average segmentation images of different methods, respectively.

be the segmentation regions of an image and $\Theta_{GT} = \{\Phi_n, n = 1, 2, \ldots, N\}$ be the true regions of ground truth map. For each region $R \in \Theta$ and $\Phi \in \Theta_{GT}$, the overlap rate can be denoted by

$$\varphi(R, \Phi) = \frac{|R \cap \Phi|}{|R \cup \Phi|} \tag{24}$$

where $|\cdot|$ denotes the pixel number of regions. For each region $R \in \Theta$, a match pair $\{R, \Phi\}$ can be obtained if the overlap rate of region Φ and region R is maximal among all of the regions $\Phi_n \in \Theta_{GT}$. Then, the SCC between Θ and Θ_{GT} can be defined as

$$\mathrm{SCC}(\Theta, \Theta_{GT}) = \frac{1}{|I|} \sum_{R \in \Theta} |R| \cdot \max_{\Phi \in \Theta_{GT}} \{\varphi(R, \Phi)\}. \tag{25}$$

Note that an ideal segmentation, i.e., the segmentation result fully matches the ground truth, SCC should be 1. The better the segmentation is, the closer to 1 the SCC is. The VI measures the distance between two segmentations in terms of their average conditional entropy given by

$$VI(R, \Phi) = H(R) + H(\Phi) - 2Mu(R, \Phi) \tag{26}$$

where H and Mu represent the entropies and mutual information between segmentations and ground truth.

Table II presents the quantitative evaluation measures of different methods for the synthetic SAR image with four-look and two-look Gamma speckle noise. The higher the number of looks, the more severe the speckle noise. It can be seen that all the four methods can achieve satisfactory segmentation results for the synthetic SAR image with four-look speckle noise. The reason is that the speckle is not serious and the image backscattering is simple. However, in comparison with other three state-of-the-art methods, our proposed approach has the best performance. When we increase the speckle level from four-look to two-look, the performance of other three comparison methods decreases with different extents. In contrast, the segmentation accuracy of our proposed method is basically the same as before, indicating that the proposed method is robust to the speckle noise.

Fig. 14 gives the P-R curves of different methods for the four SAR data sets, where the green dashed lines denote the standard F measure curves. From Fig. 14, we can find that the proposed method performs the best among the four methods for different SAR images. In Fig. 14(a)–(c), the GIOESP has lower precision and higher recall values than the proposed method, indicating that the GIOESP has more over-segmentation errors. For the GF-3 C-band data in Fig. 14(d), the GIOESP has a similar performance as proposed method. The RCBLP method performs fine in Fig. 14(a), (c), and (d). Nevertheless, in comparison with the proposed and GIOESP methods, it has lower precision and recall values due to the over-merging and over-segmentation errors. The RTPA method has a similar performance with RCBLP in the four segmentation results.

Table III illustrates the quantitative evaluation measures of different methods for the four real SAR data sets, where the bold fonts denote the best indices. It can be found that for the CETC X-band, MiniSAR Ka-band, and ICEYE X-band images, the proposed method achieves the best values in terms of the segmentation boundary and region quality. For the GF-3 C-band image, the proposed method has comparable results as the GIOESP and outperforms other methods. From Table III, we can conclude that the segmentation regions generated by the proposed method are closer to the ground truth than other state-of-the-art methods.

Fig. 15 presents the mean and standard deviation values of the quantitative evaluation measures for different methods. From Fig. 15, it can be found that for different SAR data sets,

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TABLE II

QUANTITATIVE EVALUATION MEASURES OF DIFFERENT METHODS FOR THE SYNTHETIC SAR DATA SET

Methods	The synthetic SAR Image (4-look)					The synthetic SAR Image (2-look)				
	Precision	Recall	F-measure	SCC	VI	Precision	Recall	F-measure	SCC	VI
Proposed method	**0.94**	**0.97**	**0.95**	**0.94**	**0.76**	**0.94**	**0.96**	**0.95**	**0.93**	**0.77**
GIOESP	0.90	0.96	0.92	0.87	1.26	0.88	0.93	0.90	0.85	1.34
RCBLP	0.91	0.92	0.91	0.89	1.12	0.91	0.91	0.91	0.89	1.15
RTPA	0.92	0.95	0.93	0.91	0.94	0.90	0.91	0.90	0.89	1.02

TABLE III

QUANTITATIVE EVALUATION MEASURES OF DIFFERENT METHODS FOR THE FOUR REAL SAR DATA SETS

Methods	CETC X band			MiniSAR Ka band			ICEYE X band			GF-3 C band		
	F-measure	SCC	VI	F-measure	SCC	VI	F-measure	SCC	VI	F-measure	SCC	VI
Proposed method	**0.87**	**0.81**	**1.27**	**0.88**	**0.80**	**1.05**	**0.89**	**0.83**	**1.78**	0.85	0.79	**1.89**
GIOESP	0.85	0.78	1.86	0.85	0.73	1.67	0.85	0.75	2.23	**0.86**	**0.80**	2.01
RCBLP	0.81	0.67	2.24	0.79	0.62	2.23	0.82	0.72	2.31	0.84	0.74	2.34
RTPA	0.80	0.64	2.31	0.75	0.56	2.89	0.81	0.68	2.84	0.82	0.69	2.53

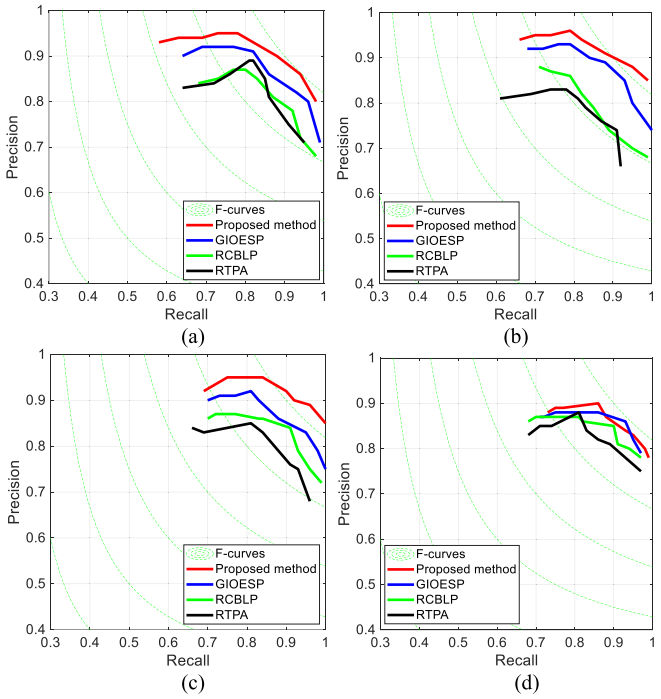


Fig. 14. P-R curves of different methods for the (a) CETC X-band data, (b) MiniSAR Ka-band data, (c) ICEYE X-band data, and (d) GF-3 C-band data.

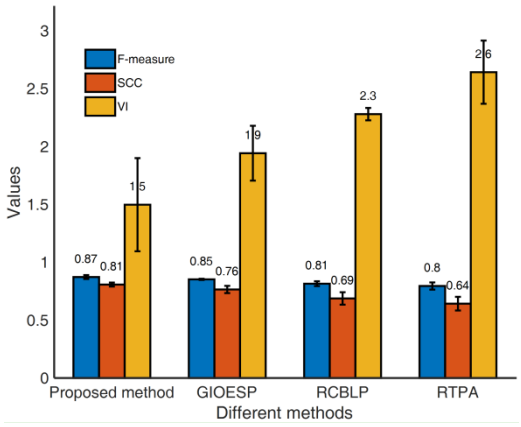


Fig. 15. Standard deviation analysis of the quantitative evaluation measures for different methods.

the F-measure of our proposed method can achieve 0.87 with a very small standard deviation, indicating that this method is robust to different types of SAR data. The SCC measure has a similar trend, exhibiting satisfactory stability for different SAR images. Although the VI measure of our proposed method is the lowest among different methods, the standard deviation is the largest. The reason is that VI values are much higher in the segmentation results of ICEYE and GF-3 data than the results of CETC and MiniSAR data. This indicates that the proposed method should be more suitable for the segmentation of high-resolution airborne SAR images than spaceborne data. The mean values of three indices of the GIOESP algorithm are worse than those of our proposed method. Nevertheless, the standard deviations of three measures for the two methods are similar. The RCBLP and RTPA have similar performance in terms of F-measure and SCC, which exhibit lower values and worse stability than other two methods.

In order to discuss the individual contributions of different components within the proposed method, Table IV gives the quantitative evaluation measures of our proposed method with different combinations for the four real SAR data sets. Considering that both the merging order and merging predicate contribute to the accuracy while the disjoint set can improve the efficiency, we discuss the merging order and merging predicate in this section and the disjoint set in Section V-D. From this table, we can find that the segmentation accuracy will decrease dramatically without the statistical dissimilarity measure, indicating that the pixel statistical information plays a key role in the definition of dissimilarity measure. The common boundary length penalty can significantly improve the segmentation accuracy due to the advantages of edge constraint. In contrast, the contribution of homogeneity constraint to the segmentation is relatively weak. In comparison with the fixed

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TABLE IV
QUANTITATIVE EVALUATION MEASURES OF THE PROPOSED METHOD WITH DIFFERENT COMBINATIONS FOR THE FOUR REAL SAR DATA SETS

Proposed methods	CETC X band			MiniSAR Ka band			ICEYE X band			GF-3 C band		
	F-measure	SCC	VI	F-measure	SCC	VI	F-measure	SCC	VI	F-measure	SCC	VI
Without statistical dissimilarity measure	0.43	0.57	3.86	0.32	0.55	4.37	0.41	0.61	4.68	0.38	0.56	4.11
Without common boundary length penalty	0.78	0.69	2.87	0.74	0.64	3.13	0.71	0.74	3.24	0.73	0.63	2.93
Without homogeneity constraint	0.84	0.74	1.86	0.83	0.77	1.77	0.82	0.80	1.94	0.82	0.76	2.01
Fixed merging threshold	0.71	0.68	2.66	0.69	0.62	2.87	0.64	0.71	3.11	0.66	0.65	2.47

TABLE V
TIME COSTS OF DIFFERENT METHODS FOR THE FOUR REAL SAR DATA SETS WITH DIFFERENT INITIAL OVER-SEGMENTATION PARTITIONS

SAR datasets	Image size (height×width)	Number of initial over-segmentation partitions	Proposed method	Proposed method without disjoint-set	GIOESP	RCBLP	RTPA
CETC X band	587×690	1500	**15.3 s**	24.2s	22.3 s	15.7 s	32.1 s
		3000	**16.2 s**	57.4s	50.2 s	28.4 s	57.7 s
MiniSAR Ka band	706×780	2000	18.2 s	29.8s	26.7 s	**17.4 s**	24.3 s
		4000	**19.3 s**	66.4s	59.5 s	30.5 s	52.4 s
ICEYE X band	890×1025	2500	**21.4 s**	43.2s	38.7 s	22.7 s	35.8 s
		5000	**22.1 s**	90.6s	70.4 s	40. 8s	62.1 s
GF-3 C band	887×1016	2500	**20.3 s**	38.7s	34.9 s	21.7 s	33.5 s
		5000	**21.8 s**	86.4s	65.1 s	39.9 s	62.3 s

merging threshold, the adaptive merging threshold can increase the segmentation accuracy of SAR images by 10%–15%, indicating that the adaptive merging threshold is beneficial to the superpixel merging and SAR image segmentation.

D. Comparison of the Computation Costs

For the four segmentation algorithms, the time cost mainly consists of initial over-segmentation partition (superpixel generation) and region (superpixel) merging. Note that the initial over-segmentation partition strategies of these four methods are all already existing, the four methods mainly focus on the region (superpixel) merging; therefore, in this subsection, we give the specific computation costs of the merging stage of different methods for the four SAR data sets and discuss the implementation efficiency. It is worth pointing out that in order to validate the contribution of disjoint set to the segmentation efficiency, we assume that other parameter settings of our proposed method are the same but do not use the disjoint sets to compare the computation time.

Table V lists the time costs of different methods for the four data sets with different initial over-segmentation partition number, where the bold fonts denote the best indices. From this table, we can see that our proposed method with the disjoint set is highly efficient for different SAR data sets, even with the increasing number of initial over-segmentation partitions. The reason is that our proposed method can map the superpixel pairs to pixel pairs thanks to the disjoint set, resulting in that the computation cost is mainly determined by the size of image. In contrast, without the disjoint set, the method runs much slower than the original proposed method due to the computation of superpixel merging order, especially when the number of superpixels is large. The time costs of GIOESP method are 50%–70% higher than those of the proposed method. Note that the main time costs come from the calculation of GIOESP for each two adjacent regions. Therefore, with the increasing number of over-segmentation partitions, the time costs increase dramatically. The RCBLP method is much faster than the GIOESP due to the utilization of the NNG for fast search minimal edge on a region adjacency graph. Therefore, the RCBLP method runs fast and is as highly efficient as the proposed method if the number of initial over-segmentation partitions is below 3000. Nevertheless, when the number is high, the time costs of RCBLP are much higher than those of proposed method. The time cost of RTPA method is basically the same as GIOESP method, which is higher than that of our proposed and the RCBLP methods. The main time consuming comes from the computation of texture pattern similarity measure with the RCBLP for each pixel pair. Overall, it can be found that although our proposed method considers multiple cues during superpixel merging, the method is still highly efficient compared with other state-of-the-art methods. For images with larger sizes, we can adjust the merging rules to achieve rapid segmentation at the sacrifice of a certain degree of segmentation accuracy.

VI. CONCLUSION

In this article, a fast superpixel region merging algorithm for SAR image segmentation is proposed. To eliminate the mixed superpixels generated by our previously proposed ALFCE, a sketch edge map is defined and used to optimize the superpixels, leading to an initial over-segmentation result for SAR imagery. Then, the SRM framework is utilized to achieve fast superpixel merging. As one of the main contributions in this article, a new merging order with the consideration of statistical dissimilarity measure and common boundary length penalty, as well as the homogeneity constraint for each superpixel pair is proposed. Therefore, the over-segmentation errors can be reduced and small man-made structures can

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also be preserved, resulting in a high segmentation accuracy. Moreover, an adaptive merging threshold according to the image complexity is defined, making the proposed superpixel merging no need to set any parameters in advance. More importantly, we used the disjoint set to map the superpixel pairs to pixel pairs for the sake of fast region merging, which forms the basis of rapid segmentation.

The proposed superpixel merging and segmentation strategies for SAR images have the advantages which are high efficiency and few parameters. It is worth pointing out that the merging rules of our proposed method can be further adjusted in order to achieve even faster segmentation for large image scenes in practical applications. Moreover, more SAR data sets with different image resolutions will be used to validate the stability of our method.

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